VISHWA KUMAR VENKATESWARAN

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SUMMARY

Security researcher and offensive security engineer with 10+ years of programming experience (C/C++, Python, Java), focused on AI/LLM security — prompt-injection testing, RAG pipeline hardening, and adversarial ML — alongside vulnerability research, exploit development, and cloud security fundamentals (AWS). Active bug bounty hunter across Bugcrowd/VDP programs with hands-on work in memory corruption, protocol-level analysis, malware/RE tooling, and offensive-security infrastructure end-to-end: recon, exploitation, PoC, and disclosure.

TECHNICAL SKILLS

- **AI & LLM Security:** Prompt Injection Testing, RAG Pipeline Security, ChromaDB (Vector DB), LangChain4j, LLM Red Teaming, Adversarial ML, Ollama, Hugging Face, TensorFlow, n8n
- **Offensive Security & Vulnerability Research:** Burp Suite, Metasploit, Ghidra, Nmap, Wireshark, YARA, Static/Dynamic Analysis, Exploit Development
- **Cloud & Infrastructure Security:** AWS (IAM, EC2, S3 Security Fundamentals) — in progress, Docker, Linux, Container Security
- **Languages & Systems:** C/C++, Python, Java, Bash, C# (.NET), Data Structures & Algorithms, Git, REST/Relational Data Flows
- **AI & Vibe Coding Tools:** Ollama, Qwen, Hugging Face, TensorFlow, LangChain4j, OpenClaw, Hermes, Cursor, Codex, Replit, Lovable

EXPERIENCE

Bug Bounty & Vulnerability Researcher — Bugcrowd *(Remote)* *May 2026 – Present*

- Perform independent vulnerability research on production web applications, identifying and responsibly disclosing security flaws through Bug Bounty and VDP channels.
- Design and run structured reconnaissance and manual testing methodologies to validate findings before submission, reducing false-positive reports.
- Document root-cause analysis for each finding, following coordinated disclosure practices used across enterprise security teams.

Summer Intern — Wimera Systems, Bengaluru *(Onsite)* *May 2026 – Jun 2026*

- Built and validated Modbus RTU and TCP/IP communication layers between industrial devices and software systems, supporting 32+ concurrent device connections with <200ms latency.
- Shipped 2+ C# (.NET) tools for protocol validation and device communication testing, cutting manual troubleshooting time for the team.
- Debugged 50+ protocol/system-level failures across a distributed set of connected devices, strengthening reliability of the communication stack.

Machine Learning Intern — SaiKet Systems *(Remote)* *Jan 2026 – Feb 2026*

- Applied ML techniques for classification and anomaly detection on security-related datasets, including data preprocessing and model optimization.
- Documented experimental workflows for reproducibility, a practice carried into later independent AI security research.
- Analyzed security-related datasets and tuned ML classification and anomaly-detection workflows to improve detection accuracy.

Security Intern — InLighnX Global Pvt. Ltd. *(Remote)* *Oct 2025 – Dec 2025*

- Completed structured training in offensive and defensive security methodologies, then built Python automation tools for authentication testing and file protection.
- Applied hashing (MD5/SHA) and password-cracking techniques to evaluate real-world credential handling weaknesses.

PROJECTS

Multi-Agent Banking Security Assistant *(AI Security)* *Dec 2025 – Feb 2026*

- Architected a privacy-preserving multi-agent banking assistant using LangChain4j, separating intent classification from privileged banking operations via a locked-down LLM-to-API tool-calling boundary.
- Built a RAG pipeline over ChromaDB with local LLaMA 3 inference (Ollama) to minimize exposure of sensitive financial data to third-party AI services.
- Red-teamed the multi-agent pipeline against prompt-injection, data-exfiltration, and tool-calling abuse scenarios, hardening guardrails between agents.

USB Rubber Ducky — HID Attack Simulation *Jun 2026 – Aug 2026*

- Developed a Raspberry Pi Pico-based USB HID attack simulation platform to assess endpoint exposure to malicious USB devices.
- Simulated keystroke-injection attacks in an isolated lab environment to evaluate endpoint security controls and attack detection capabilities.
- Conducted security assessments focused on unauthorized HID devices, USB-based initial access, and endpoint hardening.

Phishing Simulation Toolkit *Jan 2026 – Apr 2026*

- Built a phishing simulation toolkit for controlled security research and awareness testing, with tunnel-based hosting and customizable web templates.
- Researched browser permission handling and client-side attack surfaces to identify realistic social-engineering vectors.
- Automated phishing campaign setup and response tracking to evaluate user interaction and security awareness.

ACHIEVEMENTS & RECOGNITION

- Certified: Android Bug Bounty Hunting (EC-Council); BASH Training (IIT Bombay).
- Active CTF competitor — 10+ HackTheBox machines completed; top 15% on TryHackMe.
- Publishes technical security research and CTF writeups: vishwakumarv.github.io/writeups

EDUCATION

PSG College of Technology, Coimbatore *Aug 2023 – May 2027 (Expected)*

Bachelor of Engineering in Electronics and Communication Engineering

Languages: Tamil (Native), English (Professional), Hindi (Professional), Kannada (Limited)